

ORIGINAL ARTICLE

Drying kinetic and artificial neural network modeling of mushroom drying process in microwave-hot air dryer

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Abstract

Modeling of moisture content variation under variable microwave power in microwave-hot air dryer is challenging. In this study, one static and one dynamic of ANN were investigated for modeling of whole mushroom drying process. The experiments were done in three levels of hot air temperature (23, 50, and 70 °C), three levels of microwave power density (MPD) (1.5, 2, and 2.5 W/g) and two statuses of microwave power during drying process (constant and variable). The results demonstrated that the two distinct falling rate periods were observed at low MPD with any air temperatures and power density status. The lowest specific energy consumption was found in 23 °C and 2.5 fixed MPD and lowest color deterioration found in 70 °C and 1.5 fixed MPD. The results also showed the 3–6–7–1 structure of dynamic model with 0.2179 RMSE and 0.9914 *R* values had better results than other structures and would be very suitable in a predictive control system for drying control.

Practical applications

Volumetric and high efficiency heating are the primary benefit of microwave drying and lot of industries and producers interest to use it. Also, modeling of drying kinetics and evaluating of product quality in each drying method provide valuable information to improve their drawbacks and recommend it for industry. Application of microwave technology in dehydration of mushroom can result in significant improvement in quality and energy consumption. On the other hand investigating a robust ANN model for predicting the moisture content helps to better prediction and control of microwave-hot air dryer in industrial scale.

1 | INTRODUCTION

Fresh mushrooms are one of the extraordinarily perishable and its shelf life is only about 24 hr at ambient conditions (Giri & Prasad, 2007). Various physiological and morphological changes occur after harvest even at the refrigerator conditions, which make these mushrooms unacceptable for consumption. Accordingly, they should be consumed or processed promptly after harvest and for this reason the mushrooms are traded mostly in processed form in the world market.

Drying is the oldest method of preserving foods. Principle of drying process is based on the removal of water from material by means of evaporation for longer shelf life and minimizing packaging requirements. For food drying materials, several drying technologies such as hot air drying, microwave drying, vacuum drying, infrared drying technique, and their combinations have been investigated. Hot air drying

process has some drawbacks like low energy efficiency, long drying time and decrease the quality attributes but then again it has some advantages like removal surface water very well and low operating cost. Microwave drying compared with conventional hot air drying has lot of benefits to apply in food drying process like volumetric heating, high thermal efficiency, shorter drying time, and improved product quality. Therefore, for using the advantages of these two sources of energy for drying, a lot of researchers combined them together. The some newest studies for evaluating the drying kinetic of apple slices (Horuz, Bozkurt, Karataş, & Maskan, 2018) and raspberries (Szadzinska, Lechtanska, Pashminehazar, Kharaghani, & Tsotsas, 2018) were done in the hybrid microwave-hot air dryer for demonstrating its benefits rather than other drying methods.

In conventional microwave drying method, a fixed microwave power was used during the drying process. But the water of the

product evaporated and mass of product decreased over the time that resulted microwave power density (MPD) increased during the drying process. The MPD (W/g) was defined as the power to be applied to the product per unit mass of product. Increasing the power density, especially in the end of process, represents that the microwave energy is higher than required energy to evaporate of water. The excess energy absorbed by product and converted to heat energy and sharply increased the product temperature (Kone et al., 2013; Li, Raghavan, & Orsat, 2010). High temperature of products led to the deterioration of the product quality and a lot of approaches have been investigated for controlling the product temperature based on microwave power controlling during the drying process. One of them was variable power program for microwave drying process and earliest has been investigated by Shivhare, Raghavan, and Bosisio (1991). Later, many researches used this method for preventing the risk of overheating and charring of product (Botha, Oliveira, & Ahne, 2012; Clary, Mejia-Meza, Wang, & Petrucci, 2007; Clary, Wang, & Petrucci, 2005; Pereira, Marsaioli, & Ahne, 2007; Setiady, Clary, Younce, & Rasco, 2007; Wang & Xi, 2005). Also assessing the drying process under constant and variable microwave power was done for banana as a heat sensitive food and resulted the drying at variable microwave power was more suitable for drying (Ahne, Pereira, Staack, & Floberg, 2007). They also mentioned that variable microwave power can reduce charring of the product. Also the obtained results by Li et al. (2010) illustrated that the power requirements for the drying process with fix product temperature could be divided into three stages: a power increasing, a power decreasing, and a maintained power stage. Also their results indicated that power requirements are mostly related to the moisture content of the samples.

Drying is a complex process because of its simultaneous heat, mass and momentum transfer phenomena and complexity further increases if any time varying of drying conditions applies during drying process. On the other hand, drying process is a dynamic process that it demonstrated the next status of drying process depended to the current or past status. Therefore, the effective models for evaluating the effect of process parameters, optimizing of the drying process, energy integration, and control are necessary (Kumar, Karim, & Joarder, 2014). The development of mathematical and numerical models to describe the drying processes has been the topic of many research studies for several decades (Hernandez-Perez, Garcia-Alvarado, Trystram, & Heyd, 2003). In the mathematical models, some assumptions are obvious to develop (Kumar et al., 2014) and numerical models require deep knowledge of the process mechanism, estimation of many experimental parameters and application of advanced calculation methods. In order to overcome these modeling difficulties and accurately predict the moisture content with time varying drying condition during drying process, the black-box modeling approach can be investigated.

Artificial neural network (ANN) is one of the black-box modeling approaches has superiority as compared with other conventional modeling method. The main advantages of ANN are easy definition of non-linear and complex systems, short computing time and adaptive performance (Bai, Xiao, Ma, & Zhou, 2018; Kaminski & Tomczak, 1999). Also the ability of ANN for the modeling of drying process was comprehensively discussed by Farkas (2013) and an excellent

review about it was done by Aghbashlo, Hosseinpour, and Mujumdar (2015). In another study, the drying kinetics of carrot cubes with dimensions of 4, 7, and 10 mm through fluidized bed dryers were predicted by using static and dynamic neural networks. The inlet air temperatures were 50, 60, and 70 °C, and the depths were 3, 6, and 9 cm. At first, the static ANN was applied and its outputs (relative humidity and the speed of drying) were related to four exogenous inputs (drying time, drying temperature, the size of the carrot cubes, and the depth of the bed). In dynamic/recurrent ANNs, in addition to four external inputs, the input and output states were applied (relative humidity or speed drying). The obtained results expressed that recurrent ANN compared with static ANN had higher values of R^2 in prediction (Nazghelichi, Aghbashlo, Kianmehr, & Omid, 2011). Also Poonnoy, Tansakul, and Chinnan (2007) results revealed that the dynamic ANN had good ability for moisture content and temperature modeling in microwave-vacuum drying process. Inputs for the ANN models were drying time ($t_i + 1$), initial temperature (T_0) and moisture content (MC_0), microwave power, and vacuum pressure. The outputs were temperature ($T_i + 1$) and moisture content ($MC_i + 1$) at a given $t_i + 1$.

The literature review demonstrated that variable microwave power was suitable for applying in microwave-hot air, but there is still lack of effective model for predicting the moisture content during drying process with variable microwave power application and controlling the microwave power based on it. Therefore, in this study, the drying kinetic study of whole mushroom in the hybrid microwave-hot air dryer under constant and variable MPD was done and also robust static and dynamic ANN models for predicting the moisture content was developed.

2 | MATERIALS AND METHODS

2.1 | Materials and drying equipment

Button Mushrooms (*Agaricus bisporus*) were used in this study and the AOAC (1980) method was applied to determine the initial moisture content of mushrooms and its value obtained 89.65% based on wet basis.

The microwave-hot air prototype drier was designed in Uremia University (Hazervazifeh, Moghaddam, & Nikbakht, 2017). The dryer has a maximal output power of 1,000 W operating at a frequency of 2,450 MHz, as per the magnetron specification. The drying chamber had 40 × 37 × 23 cm dimensions and drying tray area is suitable for about 0.5 kg of product. For overcome the no uniform distribution of electromagnetic energy in microwave cavity, drying tray was rotated with 4 rpm. Product mass (accuracy of ±1 g) was recorded by a load cell at 3 min of time interval.

In microwave-hot air dryer, the hot-air temperature and velocity, sample thickness and MPD are the effective parameters on mushroom drying kinetic. Air temperature was set up and controlled by a separate panel in three levels: 23 (environment temperature), 50, and 70 °C. The inlet air flow was located in the middle of the chamber and distributed in the cavity with 0.25 m/s of velocity in all experiments. The air flow was perpendicular with the sample surface. In this study,

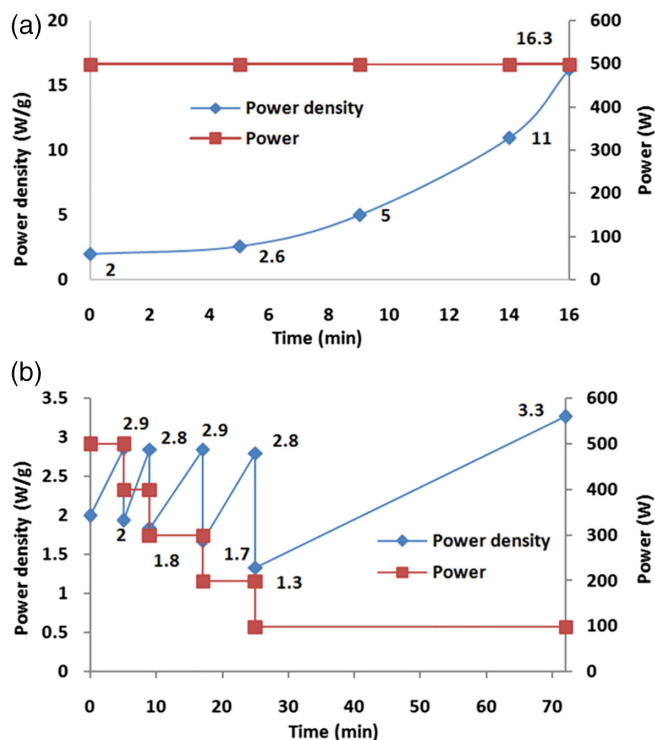


FIGURE 1 Variations of microwave power and power density with time: (a) variable power density, (b) almost constant power density, ($T = 70^\circ\text{C}$ and initial microwave power density 2 W/g)

the mushrooms with same diameter ($4 \pm 0.15\text{ cm}$) and color were selected and located on tray in the dryer.

The MPD was applied in two statuses: variable MPD and almost constant MPD (Figure 1). In the variable power density, microwave power maintained constant during drying process and with decreasing the moisture content of product, the power density was increased (Figure 1a). In the constant power density with removing of the moisture content, the microwave power decreased to maintain the constant MPD. In this method during the drying process, microwave power was reduced by 20% with concerning the moisture content of product for preventing to reach the high power density (Figure 1b). In both method the drying process started with three levels of MPD of 1.5, 2, and 2.5 W/g . All experiments started with 500 W of microwave power and for obtaining to 1.5, 2, and 2.5 W/g in each experiment 333, 250, and 200 g of mushroom were used, respectively.

TABLE 1 Uncertainties of the parameters during drying experiment

Description	Unit	Estimated uncertainty
Mass loss measurement	g	± 1.118
Moisture quantity measurement	g	± 0.0141
Drying time measurement	min	± 0.141
Drying air temperature measurement	$^\circ\text{C}$	± 0.02
Drying air velocity measurement	m/s	± 0.17

TABLE 2 Dynamic models of the ANN for modeling of moisture content

Model type	Model number	Inputs model	Output model
Static	SANN	$T, P/\text{MC}, t$	$\text{MC}(t)$
Dynamic	DANN	$T, P/\text{MC}, \text{MC}(t)$	$\text{MC}(t + \Delta t)$

Drying rate (DR) and moisture ratio (MR) are most important factors of drying kinetic and study of these factors revealed the good information about drying process. The MR and DR calculated by the following equations (Hazervazifeh et al., 2017):

$$\text{MR} = \frac{\text{MC}(t)}{\text{MC}_0} \quad (1)$$

$$\text{DR} = \frac{\text{MC}(t + \Delta t) - \text{MC}(t)}{\Delta t} \quad (2)$$

where, $\text{MC}(t)$ is the moisture content at time t and MC_0 is the initial moisture content, $\text{MC}(t + \Delta t)$ moisture content at time $t + \Delta t$, Δt is the time interval to measure weight of product during drying process. In these formulas, the dimension of Δt , DR, and MC are min, kg/kg min (d.b.), and kg/kg (d.b.), respectively.

To have an exact insight of two status of MPD method in the processing of mushroom drying, specific energy consumption (SEC), and color deterioration were monitored and analyzed in this research. The first benefit of this analysis is better judgment which status of microwave-hot air dehydration in food industry and scaled-up projects would be applicability. In this study, a digital watt-meter was connected to high voltage circuit as a power supply of magnetron lamp, heaters and blowers for measuring the energy consumption. Then energy required for evaporating 1 kg of water from the sample, specific energy consumption, was calculated by the following Equation (3):

$$\text{SEC} = \frac{E_t}{m_{\text{eva}}} \quad (3)$$

where, E_t is total energy consumption (MJ) and m_{eva} (kg) is the mass of water removed during drying process.

Color measuring of fresh and dried mushroom samples has been done by investigated machine vision system and image processing algorithm (Sharifian, Modarres-Motlagh, Komarizade, & Nikbakht, 2013). The total color difference (ΔE) and browning index (BI) were used as quality index and calculated with the following equations (Maskan, 2000):

$$\Delta E = \sqrt{(L_0^* - L^*)^2 + (a_0^* - a^*)^2 + (b_0^* - b^*)^2} \quad (4)$$

$$\text{BI} = \frac{[100(x - 0.31)]}{0.17} \quad (5)$$

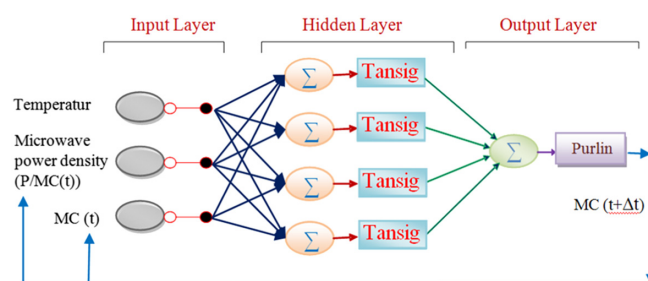


FIGURE 2 Structure of dynamic ANN

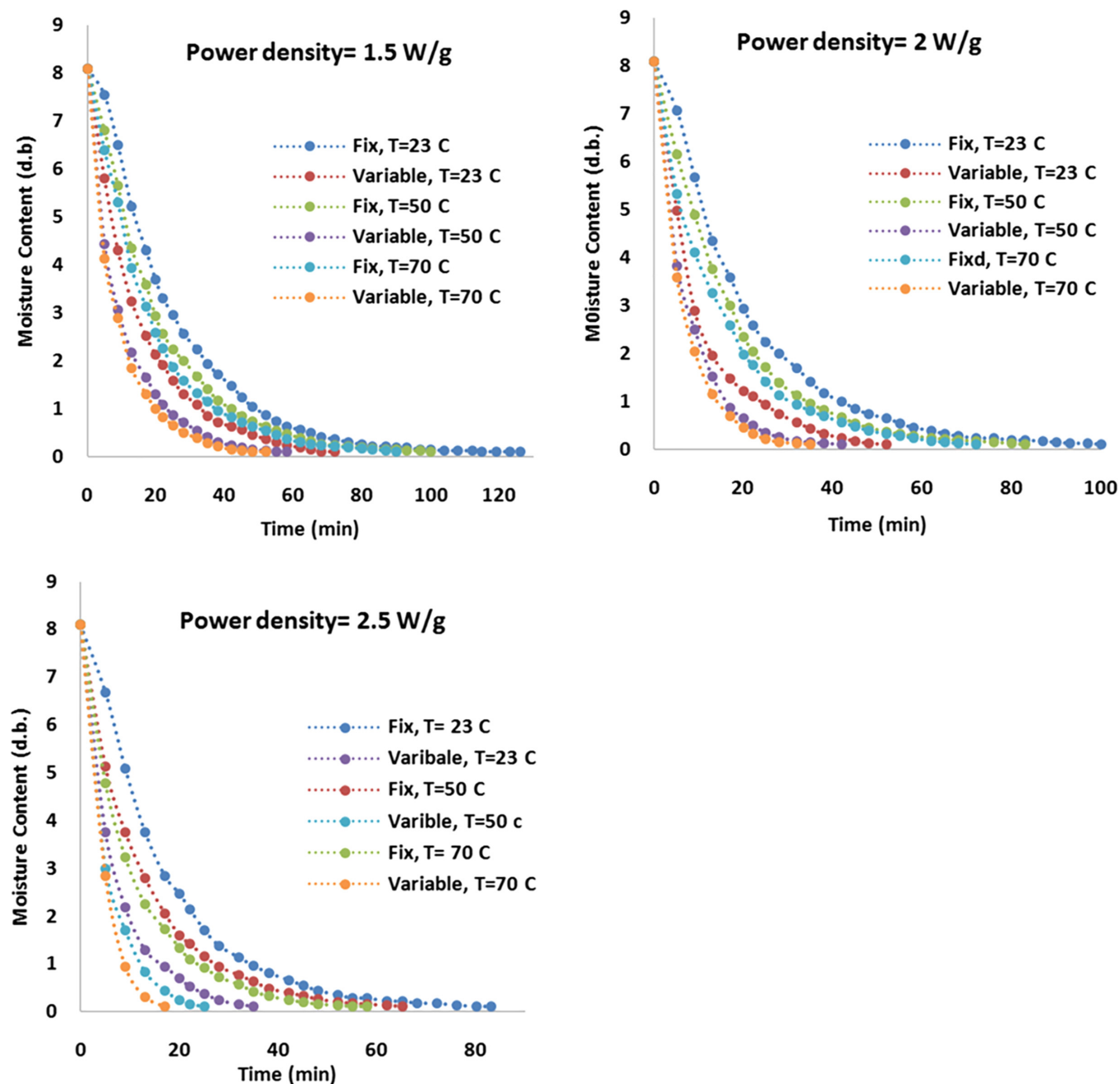


FIGURE 3 Moisture content variations during drying time with fix and variable power density in different drying temperature

$$x = \frac{(a^* + 1.75L^*)}{(5.64L^* + a^* - 3.012b^*)} \quad (6)$$

where, L^* , a^* , and b^* indicate the brightness, redness, and yellowness of dried samples, respectively, and subscript "0" refers to the color reading of fresh mushroom (Maskan, 2000).

2.2 | Experimental uncertainty

Errors and uncertainties in experiments can arise from instrument selection, condition, calibration, environment, observation and reading, and test planning (Akpınar & Bicer, 2008). Uncertainty of any measurement can be evaluated by using the following equation (Akpınar & Bicer, 2008):

$$W_R = \left(\left(\frac{\partial R}{\partial x_1} w_1 \right)^2 + \left(\frac{\partial R}{\partial x_2} w_2 \right)^2 + \dots + \left(\frac{\partial R}{\partial x_n} w_n \right)^2 \right)^{0.5} \quad (7)$$

where, W_R is uncertainty in the result, R is result as a function of independent variables of $x_1, x_2, \dots, x_i, \dots, x_n$ and $w_1, w_2, \dots, w_{xi}, \dots, w_n$ are uncertainties of independent variables. In Equation (7), the same odds are given to the all uncertainties of the independent variables. In microwave- hot air drying experiments, the hot air temperature and velocity, weight losses and moisture content were measured with appropriate instruments. Akpınar and Bicer (2008) method was used for calculating the total uncertainty in the measurement of these parameters. During the experiments, the total uncertainties of the measured parameters were presented in Table 1.

TABLE 3 Specific energy consumption and color parameters

Power density status	Temperature	Power density	Drying time (min)	SEC (MJ/kg)	ΔE	BI
Variable	23	1.5	72	8.85	31.50	118.5
		2	52	7.55	35.25	140.0
		2.5	35	6.29	41.08	197.1
	50	1.5	58	18.25	18.82	66.92
		2	42	15.19	22.42	80.81
		2.5	25	11.42	30.29	111.9
	70	1.5	52	16.5	15.21	56.60
		2	35	13.17	26.14	97.20
		2.5	17	9.35	32.77	129.8
Fixed	23	1.5	127	6.01	23.50	84.00
		2	102	5.60	26.80	97.50
		2.5	83	5.13	31.27	111.7
	50	1.5	100	26.29	15.37	57.03
		2	84	23.20	19.95	71.29
		2.5	65	20.40	21.62	74.80
	70	1.5	90	25.40	10.10	42.32
		2	71	22.41	13.88	52.41
		2.5	58	19.37	14.97	54.97

2.3 | Artificial neural network model

The ANN model has commonly three layers: input layer, one/two hidden layer(s), and an output layer. In an ANN model, selection of a suitable number of hidden layers, number of neurons in each hidden layers, activation function, and training algorithm is very important for reaching a satisfactory prediction. The trial and error method was used for finding and selecting the best topology of the ANN. A tang sigmoid function (Equation (8)) commonly was used as transfer function in the hidden layers (Lertworasirikul & Tipsuwan, 2008) of the network, and a linear function (Equation (9)) was used in the second layer. Also The Levenberg–Marquardt back-propagation algorithm with early stopping method for improving the generalization of the ANN ordinarily was used for training of the ANN model (Hernandez-Perez, Garcia-Alvarado, Trystram & Heyd, 2004).

$$\text{Tansig}(n) = \frac{2}{(1 + e^{-2n})} - 1 \quad (8)$$

$$\text{Purelin}(n) = n \quad (9)$$

The performance of ANN model was measured by root mean square error (RMSE) and correlation coefficient (R) between the predicted values of the network and the target or experimental values, as shown by Equations (10) and (11):

$$\text{RMSE} = \sqrt{\frac{\sum_{j=1}^N (z_j - o_j)^2}{N}} \quad (10)$$

$$R = \sqrt{1 - \frac{\sum_{j=1}^N (z_j - o_j)^2}{\sum_{j=1}^N (z_j - o_m)^2}} \quad (11)$$

where, z_j is the predicted output from observation j , o_j is the experimental (target) output from observation j , o_m is the average value of experimental output, and N is the total number of data observation. On the other hand, the performance of any structure of the

ANN depends on the initial weights, therefore, every structure of the ANN retrained five times and the best one selected.

In this study, the data sets obtained from experiments were divided into three sets; training (70%), validation (15%), and testing (15%). In this article, one static and one dynamic of the ANN model were investigated for predicting of the MC during the hot-air drying process of mushroom (Table 2). In the first static model, the hot air temperature (T), microwave power density (P/MC), and time (t) as input and $MC(t)$ was used as output. In the second dynamic model, the hot air temperature (T), microwave power density (P/MC), and moisture content at time t ($MC(t)$) as input and $MC(t + \Delta t)$ was used as output (Figure 2). Because of measuring of weight product and adjust the power of microwave for controlling the power density during drying process is not suitable. Therefore, instead of the P/W we used P/MC . The MC value can be measure by sensor or predict by modeling of drying process and, therefore, adjusting the microwave power based on MC value could be suitable and practical (Li et al., 2010).

3 | RESULTS AND DISCUSSION

Variation of moisture content with drying time was evaluated during the drying process with different power density, status, and hot air temperature (Figure 3). As indicated in this figure and Table 3, with increasing the power density and air temperature, the drying time decreased. For example, with air temperature of 23 °C, increasing the power density from 1.5 to 2.5 W/g led to decrease in the drying time from 127 to 83 with fixed status and 72–35 min with variable status. Also, with MPD of 2 W/g, increasing the air temperature from 23 to 50 and 70 °C reduced drying time from 52 to 42 and 35 min, respectively, with variable power density status and 102–84 and 71 min, respectively, with fixed power density status. With increasing

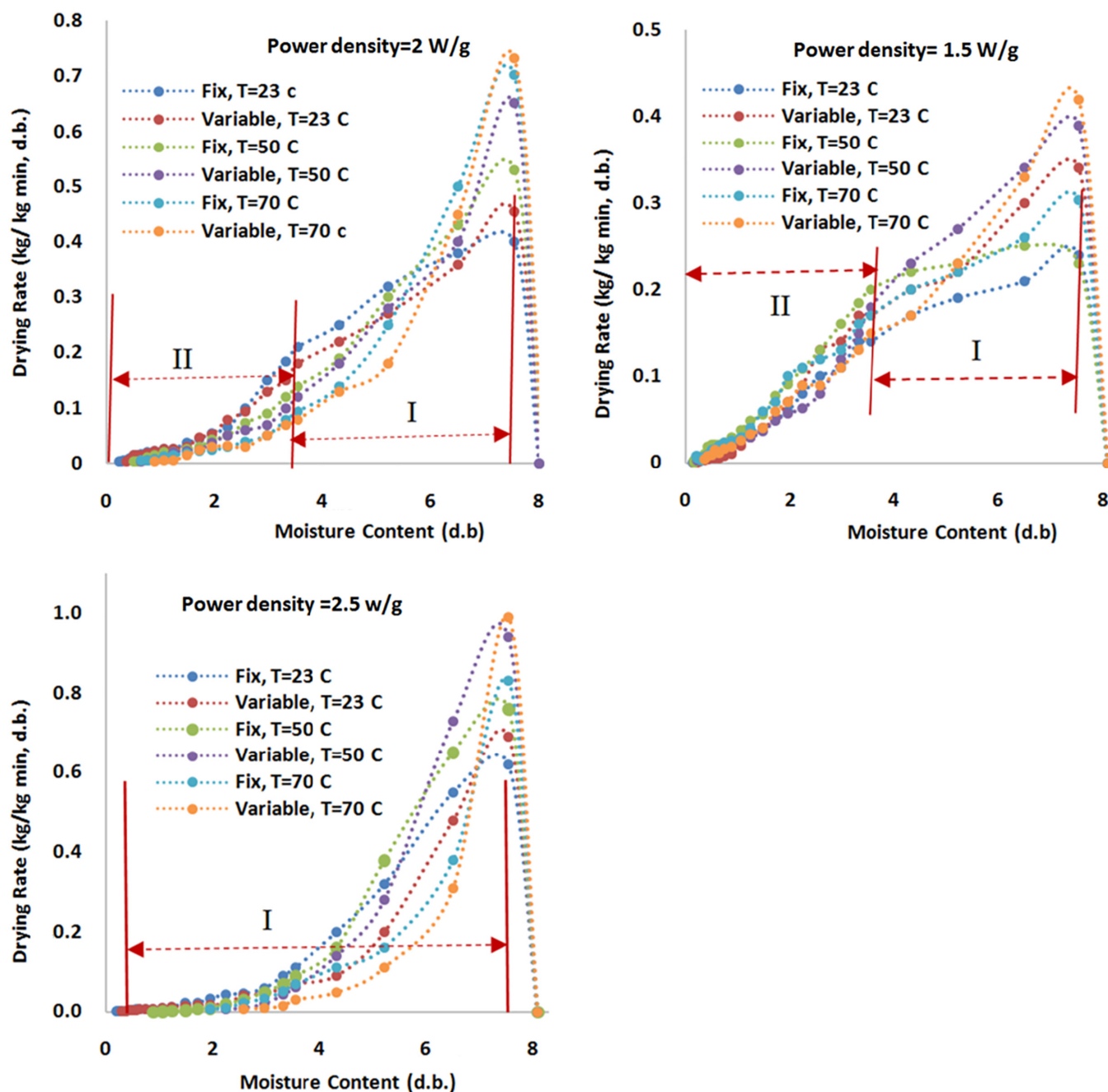


FIGURE 4 Drying rate variation versus moisture content, (I) first drying rate period, (II) second drying rate period

microwave power density the absorption of microwave energy with product and temperature gradient into the product increased and accelerated the moving of moisture content inward to the surface. Also results indicated that in MPD, drying time is reduced by the air temperature. This is because increased air temperature reduces surface mass transfer resistance and moisture content is released more quickly. Also in MPD and air temperature, drying time is increased by changing the power density status from variable to fix. In variable MPD status, the power density with removing the moisture content of product during drying process was increased exponentially. For instance, in Figure 1a, drying process started with 2 W/g but in the end of drying process it reached 16 W/g in comparison with 3.45 W/g with fixed MPD. Therefore, drying time with variable power density was lower than fix status. For example, in 1.5 W/g, by

changing the power density from variable to fix status at the air temperature of 23 °C, drying time increases from 72 to 127 min and with 2.5 W/g at the air temperature of 70 °C, it raises from 17 to 58 min. However increasing the power density during drying process had benefit for time decreasing but then again has some drawbacks like quality deterioration of food dried product. The results of this study were similar to previously reported results (Kone et al., 2013; Li et al., 2010; Pereira et al., 2007).

Variation of drying rate with time in different treatments of MPD and air temperature is given in Figure 4. Sharply increased drying rate in early stages of all experiments is considerable because the initial moisture content of the product is high and moisture loss rate increases as the products heats up gradually. Increased drying rate happens simultaneously with decreasing moisture content of the

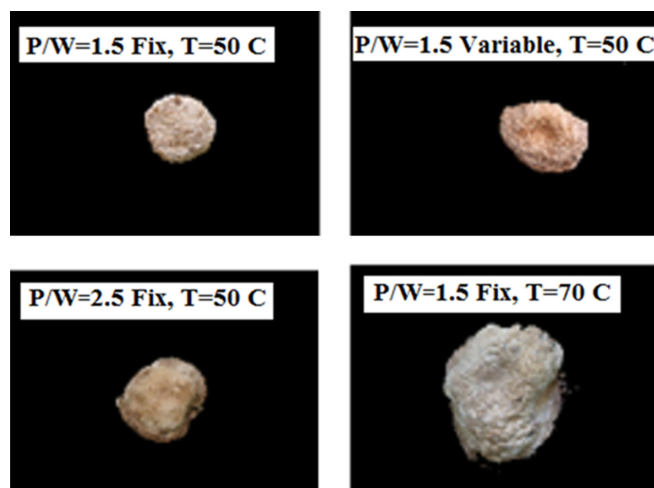


FIGURE 5 Color change after drying process

product until the product surface dried up. Thus, removing the surface moisture of product and moisture and vapor diffuse from inner layers of product to surface causes the drying rate instantly enters falling phase after it reaches its peak. The Figure 4 also demonstrated that the power density had greatest effect on the raising drying rate stage. With 1.5, 2, and 2.5 power density, the peak of drying rate located between 0.23–0.42, 0.4–0.73, and 0.76–1, respectively. The same results were obtained by Hazervazifeh et al. (2017) for apple slice and in contrast with this study, Esturk (2012) found constant rate period for Sage leaves because of the leaves have big surface than mushroom and apple slice for mass transforming. As a consequence the drying rate curves with 1.5 power density in any air temperature and 2 power density with 23 temperature after raising stage displayed two distinct falling rate periods (I and II in Figure 4). The first located from 7.5 to 3.5 of moisture contents, the second at moisture contents less than 3.5 (d.b.). But for other drying conditions one falling rate period was observed. Some studies were found the same results about two falling

drying rate periods for apple slice and carrot cube in microwave-hot air dryer (Horuz et al., 2018; Song, Li, & Raghavan, 2017; Wang & Xi, 2005). Also results demonstrated that increasing the air temperature and varying the constant power density to variable status led to slope of trend of drying rate in falling phase increased.

Table 3 expressed the SEC and color parameters values. As can be seen in the table, the lowest SEC for drying process was found in 23 °C of ambient temperature and varied 5.13–6.01 MJ/kg for fixed and 6.29–8.85 for variable power density status. The results also indicated that increasing the power density and hot air temperature from 50 to 70 °C decreased the SEC. The same results were found by other studies (Horuz et al., 2018; Motavali, Minaei, Banakar, Ghobadian, & Darvishi, 2016). To comparison the 23 °C of air temperature which the SCE with constant PMD was lower than variable status but then again with others air temperature SEC with constant power density had higher value than the variable status. In contrast with SEC, the highest total color deterioration were happened in 23 °C and varied from 31.50 to 41.08 for ΔE and 118.5 to 197.1 for BI in variable power density and varied from 23.50 to 31.27 for ΔE and 84 to 111.71 for BI in fixed power density. The color deterioration results indicated that the drying with fixed power density had lower color changes than variable state and increasing the power density increased the color changes and increasing the air temperature decreased them (Table 3 and Figure 5). Therefore, the lowest color deterioration was found 10.10 for ΔE and 42.32 for BI with highest air temperature of 70 °C and lowest MPD of 1.5 W/g with constant status. Finally, with concerning to the Table 3 and balancing between SEC as energy consumption index and color as quality index, the drying of whole mushroom with 2.5 fixed power density and 70 °C of air temperature with 58 min of drying time, 19.37 MJ/kg of SCE, 14.97 of ΔE , and 54.97 of BI and also drying with 1.5 variable power density and 70 °C of air temperature with 52 min of drying time, 16.5 MJ/kg of SCE, 15.21 of ΔE , and 56.60 of BI were recommended as best drying conditions.

TABLE 4 Results of the static model to predict the moisture content ratio

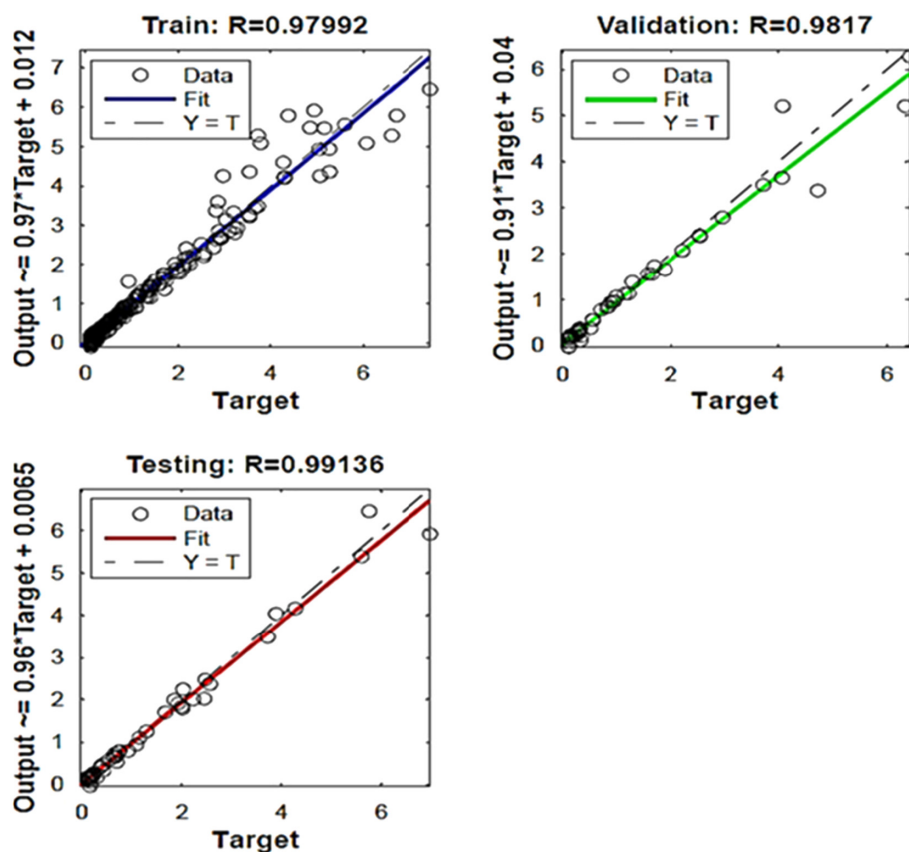
Numb of neuron in		RMSE			R ²		
First hidden layer	Second hidden layer	Train	Validation	Test	Train	Validation	Test
4	–	0.3107	0.3004	0.3996	0.9869	0.9941	0.9801
6	–	0.3146	0.3002	0.3852	0.9866	0.9942	0.9892
8	–	0.3034	0.2930	0.3503	0.9879	0.9934	0.9892
10	–	0.3221	0.2906	0.3757	0.9860	0.9945	0.9891
12	–	0.2922	0.3293	0.3822	0.9884	0.9934	0.9887
14	–	0.3063	0.3242	0.3807	0.9873	0.9932	0.9895
4	6	0.3350	0.2908	0.3978	0.9848	0.9940	0.9894
5	4	0.3326	0.2808	0.3715	0.9862	0.9940	0.9892
5	6	0.3249	0.3039	0.3853	0.9857	0.9936	0.9895
6	5	0.3354	0.2991	0.3950	0.9848	0.9938	0.9890
6	7	0.3316	0.3033	0.3899	0.9852	0.9931	0.9892
7	6	0.2961	0.3179	0.3614	0.9881	0.9935	0.9892
7	8	0.3287	0.2906	0.3705	0.9854	0.9937	0.9891
8	7	0.3133	0.2863	0.3698	0.9867	0.9944	0.9891
8	9	0.3176	0.2866	0.3861	0.9810	0.9949	0.9829
9	9	0.3066	0.3216	0.3862	0.9873	0.9934	0.9857

TABLE 5 Results of the dynamic model to predict the moisture content ratio

Numb of neuron in		RMSE			R^2		
First hidden layer	Second hidden layer	Train	Validation	Test	Train	Validation	Test
2	-	0.3754	0.3496	0.3459	0.9671	0.9787	0.9822
4	-	0.3058	0.3464	0.2455	0.9781	0.9772	0.9885
6	-	0.2787	0.3371	0.2245	0.9818	0.9785	0.9904
8	-	0.2894	0.3095	0.2264	0.9804	0.9836	0.9903
10	-	0.2982	0.3116	0.2253	0.9792	0.9857	0.9910
12	-	0.2937	0.2971	0.2568	0.9798	0.9852	0.9877
14	-	0.3747	0.3419	0.3011	0.9689	0.9799	0.9843
5	5	0.2864	0.3465	0.2219	0.9808	0.9773	0.9908
5	6	0.3083	0.3048	0.2375	0.9777	0.9871	0.9904
6	5	0.2726	0.3185	0.2396	0.9826	0.9796	0.9888
6	7	0.2943	0.3132	0.2179	0.9799	0.9817	0.9914
7	6	0.3157	0.3071	0.2453	0.9780	0.9822	0.9888
7	8	0.3050	0.3408	0.2649	0.9782	0.9821	0.9868
8	8	0.2798	0.3270	0.2253	0.9817	0.9783	0.9901
8	10	0.3680	0.2952	0.2477	0.9700	0.9837	0.9899
9	11	0.2807	0.3280	0.2447	0.9819	0.9807	0.9884

Table 4 shows the training results of the static model to predict the moisture content. As can be seen in this table, the ANN with 7 and 6 neurons in the first and second hidden layers, respectively, was elected to be the best structure for mushroom kinetic modeling. This structure had 0.2961, 0.3179, and 0.3614 of RMSE and 0.9981, 0.9935, and 0.9892 of R for train, validation, and test, respectively. In more researches (Abbaszadeh, Motevali, Khoshtaghaza, & Kazemi,

2011; Kaveh, Chayjan, & Khezri, 2018; Movagharnejad & Nikzad, 2007; Omid, Baharlooeei, & Ahmadi, 2009) drying conditions and time were be used as input in the ANN for modeling and predicting of the moisture content like the static model of this study. But this model did not have good ability for online drying control especially in microwave drying process. Because the drying process is the dynamic process and prediction of the moisture content for next step at any

**FIGURE 6** Correlation between the experimental and predicted MC values

drying conditions depends on the current moisture content and drying conditions. In the static model, there is not any input parameter for representing the current situation.

In the dynamic model (DANN) instead of the time, the moisture content at time t replaced as inputs of the ANN model and the moisture content for the next step ($t + \Delta t$) predicted by the ANN. Time was not used as input in this model and this model was a dynamic time-independent model. Table 5 shows the results of this dynamic model. As shown in Table 5, the ANN with 3–6–7–1 structure had better results than other structures. The results of this structure were 0.2943, 0.3132, and 0.2179 of RMSE and 0.9799, 0.9817, and 0.9914 of R for train, validation, and test, respectively. The R values of this model for train, validation and test were illustrated in Figure 6.

Comparing the results of static and dynamic models indicated that the dynamic model had better accuracy than static model for predicting the moisture content and would be very suitable for predictive drying control. Also the results of Li et al. (2010) demonstrated that adjustment of the microwave power based on moisture content further improved the temperature control of product than power adjustment based on the time. The advantage of proposed dynamic model in this study is predicted the moisture content for every next step and based on the predicted value of the moisture content the microwave power re-adjusted. The similar results about dynamic ANN model for predicting the temperature and moisture content of tomato slice in microwave vacuum dryer reported (Poonnoy et al., 2007). On the other hand, one of the objects of researches in food drying process was precision control of final moisture content during drying process. The same studies have been reported that the ANN had good ability for dynamic modeling of drying process and controlling of the drying process for final moisture content (Cubillos & Reyes, 2003; Behroozi-Khazaei, Tavakoli Hashjin, Ghassemian, Khoshtaghaza, & Banakar, 2013). Therefore, the developed dynamic ANN model may be used in a predictive control system, which is able to estimate the forthcoming responses of the sample when certain process control parameters are given.

4 | CONCLUSIONS

The previous studies expressed that microwave power must be controlled and adjusted based on moisture content of product, but there is still lack of appropriate model for predicting the moisture content during drying process with variable microwave power application and controlling the microwave power based on it. Therefore, in this study, the drying kinetic study of whole mushroom in the hybrid microwave-hot air dryer under constant and variable MPD was done and also robust static and dynamic ANN models for predicting the moisture content was developed. Drying kinetics showed that drying time of whole mushroom ranged from 16 to 127 min and decreased with increasing microwave power and air temperature. During the drying, two distinct falling rate periods were observed only at low microwave power and air temperature. The lowest SEC value, 5.13 MJ/kg, was found in 23 °C of ambient air temperature and 2.5 W/g of fixed power density. The lowest color deterioration was found 10.10 for ΔE and 42.32 for BI with 70 °C and 1.5 fixed power density of drying

conditions. Therefore, with concerning to the Table 3 and balancing between SEC as energy consumption index and color as quality index, the drying of whole mushroom with 2.5 fixed power density and 70 °C of air temperature with 58 min of drying time, 19.37 MJ/kg of SCE, 14.97 of ΔE , and 54.97 of BI and also with 1.5 variable power density and 70 °C of air temperature with 52 min of drying time, 16.5 MJ/kg of SCE, 15.21 of ΔE , and 56.60 of BI were recommended as best drying conditions. However, it needs to measure chemical compounds, rehydration ration, and texture properties in the future study for gain better insight into selection the best drying condition for mushroom drying. Also in this study, the ANN for static and dynamic modeling of mushroom drying process was investigated. The results revealed that the dynamic model had good accuracy compared with static model. Developing the dynamic model in this study for predicting the moisture content and adjusting the microwave power based on the moisture content would be applicable for online microwave power control.

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